

CASE STUDY REPORT

**Design and Deployment of a Highly Available,
Multi-Tier Cloud-Native Application using the
AWS Well-Architected Framework**

VeriTrust - AI-Powered Fake News Detector

Submitted by

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AWS Region: US East (N. Virginia) - us-east-1

AWS Well-Architected Review: 57/57 Questions Answered

This report clearly separates implemented AWS evidence from the proposed target architecture.

1. Executive Summary

VeriTrust is an AI-powered fake-news detection project combining a web interface with a Python/Streamlit machine-learning backend. For this AWS case study, the static frontend was deployed to Amazon S3 Static Website Hosting and verified through the S3 website endpoint. A highly available multi-tier target architecture is documented using Route 53, an Application Load Balancer, EC2 Auto Scaling across Availability Zones, Amazon RDS, VPC/security controls, IAM and CloudWatch. The target services are explicitly described as planned unless deployment evidence exists.

2. Project Overview

The application is designed to accept news article text or a public article URL and present verification-oriented analysis. The original project uses Python, Streamlit, scikit-learn, TF-IDF and supporting data/web-processing libraries. The current S3 deployment is a static presentation/demo layer; live ML prediction requires the Python/Streamlit backend.

3. Implemented AWS Deployment

Implemented and evidenced resources include an Amazon S3 bucket named `veritrust-fake-news-prashant-2026` in `us-east-1`, three frontend files (`index.html`, `style.css` and `script.js`), S3 Static Website Hosting, and successful browser access to the S3 website endpoint. The direct S3 website endpoint is HTTP; HTTPS would require a service such as CloudFront in a production design.

4. Target Highly Available Multi-Tier Architecture

The target architecture separates presentation, application and data responsibilities. Users reach the application through Route 53 and an Application Load Balancer. EC2 application instances are intended to run across multiple Availability Zones and scale horizontally. Amazon RDS is intended to provide the persistent data layer with Multi-AZ resilience. VPC, security groups, IAM and CloudWatch provide networking, access control and observability. Amazon S3 hosts the static frontend assets.

5. Multi-Tier and High-Availability Design

The architecture is designed so that a failure of an individual application instance does not take down the service. Multiple Availability Zones reduce single-AZ dependency, Auto Scaling supports changing demand, and a load balancer distributes traffic. The database tier is designed for Multi-AZ resilience. These statements describe the target architecture, not current deployment evidence.

6. AWS Well-Architected Framework Review

The VeriTrust workload was reviewed with the AWS Well-Architected Framework in the AWS Well-Architected Tool. All 57 questions were answered. The recorded review showed 16 high-risk items and 8 medium-risk items. A milestone was saved. The risk findings are treated as improvement actions rather than evidence that every recommended production control has already been implemented.

7. Six-Pillar Alignment

Operational Excellence: monitoring, documented deployment and repeatable operational processes are planned. Security: IAM least privilege, encryption, network segmentation and controlled access are emphasized. Reliability: Multi-AZ application and database design, health checks and recovery are planned. Performance Efficiency: Auto Scaling, load balancing and appropriate compute sizing are proposed. Cost Optimization: right-sizing, Auto Scaling and lifecycle controls are recommended. Sustainability: efficient

resource utilization, scaling and reduced idle capacity are recommended.

8. Deployment Evidence

The submission includes screenshots showing S3 upload success, Static Website Hosting configuration, the working VeriTrust website, the Well-Architected Tool review at 57/57 questions, and a saved milestone. These screenshots are the primary evidence for the implemented AWS portion of the case study.

9. Implementation Roadmap

Next production steps are to containerize or deploy the Python/Streamlit backend on EC2 or an equivalent compute platform, place it behind an Application Load Balancer, configure Auto Scaling across at least two Availability Zones, add a resilient database layer where persistence is required, introduce CloudFront/HTTPS for secure frontend delivery, and connect CloudWatch monitoring and alerting. Infrastructure as Code should then be added for repeatability.

10. Conclusion

VeriTrust demonstrates a clear path from a working static AWS deployment to a highly available, multi-tier cloud-native design. The submission intentionally distinguishes what is deployed and evidenced today from what is proposed as the target architecture. This keeps the case study technically honest while showing how the AWS Well-Architected Framework can guide production hardening.

Appendix A - Well-Architected Pillar Completion

Pillar	Answered
Operational Excellence	11/11
Security	11/11
Reliability	13/13
Performance Efficiency	5/5
Cost Optimization	11/11
Sustainability	6/6
Total	57/57

Appendix B - Submission Status

Implemented: S3 static frontend and its verification evidence. Target architecture: Route 53, ALB, EC2 Auto Scaling, RDS Multi-AZ, VPC/security controls, IAM and CloudWatch. Official AWS-generated WAT PDF: pending/not included in this package.